

Standard Form & Indices

Quick reference sheet · GCSE Maths, Number · keep handy for revision

1. Laws of indices

Multiplying (same base): $a^m \times a^n = a^{m+n}$ — add the powers.

Dividing (same base): $a^m \div a^n = a^{m-n}$ — subtract the powers.

Power of a power: $(a^m)^n = a^{m \times n}$ — multiply the powers.

These laws only apply when the base is the same on both sides.

e.g. $3^4 \times 3^2 = 3^6$ · $5^7 \div 5^3 = 5^4$ · $(2^3)^2 = 2^6 = 64$

2. Powers to know by heart, and estimating roots

Recall (no calculator): powers of 2 up to $2^{10}=1024$, powers of 3 up to $3^5=243$, powers of 4 up to $4^4=256$, powers of 5 up to $5^4=625$, and square numbers up to 15×15 .

Also know: $1000 = 10^3$, $1,000,000 = 10^6$.

Estimating a root: find the two whole numbers either side. e.g. $\sqrt{50}$: $7^2=49$ and $8^2=64$, so $\sqrt{50}$ is between **7 and 8**.

3. Zero and negative indices

$a^0 = 1$ for any non-zero a . e.g. $9^0 = 1$, $100^0 = 1$.

$a^{-n} = 1 \div a^n$ — a negative index flips the number into a fraction, it does not make the answer negative.

Expression	Value
7^0	1
3^{-1}	$1/3$
4^{-2}	$1/16$
10^{-2}	0.01

4. Laws of indices with algebra

The same three laws from Section 1 apply when the base is a letter. If there is a coefficient (number in front), handle the numbers and letters separately.

e.g. $x^4 \times x^3 = x^7$ · $x^9 \div x^2 = x^7$ · $(x^2)^5 = x^{10}$

e.g. $2x^3 \times 3x^4 = 6x^7$ · $(2x^3)^2 = 2^2 \times x^{3 \times 2} = 4x^6$

$x^0 = 1$, and $x^{-2} = 1 \div x^2$ (same rules as Section 3).

5. Fractional indices (Higher)

A fractional index means a root: the denominator says which root to take, the numerator is a power applied after.

$$a^{1/n} = \sqrt[n]{a} \cdot a^{m/n} = (\sqrt[n]{a})^m$$

e.g. $8^{1/3}$ = cube root of 8 = **2**. $4^{3/2} = (\sqrt{4})^3 = 2^3 = \mathbf{8}$. $9^{-1/2} = 1 \div \sqrt{9} = \mathbf{1/3}$.

6. Standard form: writing large and small numbers

Standard form writes any number as $A \times 10^n$, where $1 \leq A < 10$ and n is an integer. Large numbers (10 or more) use a positive power; small numbers (less than 1) use a negative power.

Ordinary number	Standard form
8,000	8×10^3
450,000	4.5×10^5
0.006	6×10^{-3}
0.00072	7.2×10^{-4}

7. Calculating with standard form

Multiply: $(A \times 10^m) \times (B \times 10^n) = (A \times B) \times 10^{m+n}$

Divide: $(A \times 10^m) \div (B \times 10^n) = (A \div B) \times 10^{m-n}$

If the front numbers multiply to 10 or more, adjust back into standard form. e.g. $(5 \times 10^6) \times (4 \times 10^3) = 20 \times 10^9 = \mathbf{2 \times 10^{10}}$.